



BASIC EMERGENCY CARE

Envenomation

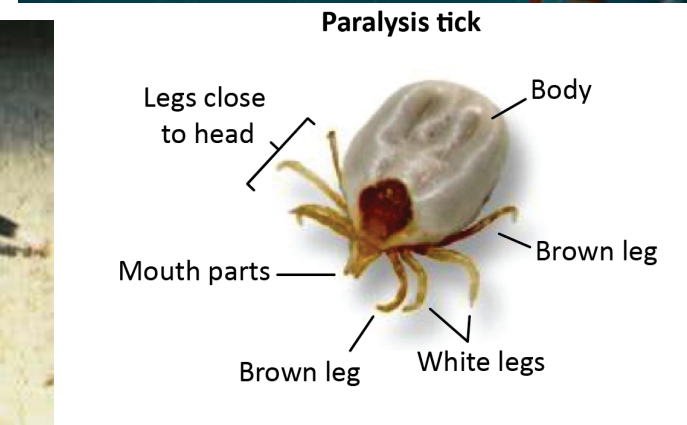
Dr Nishtiman Mohammed

ENVENOMATION

- ❓ Envenomation occurs when a venomous animal injects sufficient venom by a bite or a sting into a prey item or perceived predator to cause deleterious local and/or systemic effects. (Venom-induced disease)
- ❓ VENOM is a complex mixture of diverse components (notably toxins), often with several separate toxins that can cause adverse effects in humans , and each potentially capable of multiple effects.
- ❓ Two types of bite : dry and wet bite.

Venomous animals and human envenoming

Phyla	Principle venomous animal group
Chordata	Snakes Spiny fish Stingrays
Arthropoda	Scorpions Spiders Paralysis ticks Insects
Mollusca	Cone snails Blue-ringed octopus
Coelenterata	Jelly fish



KEY VENOM EFFECTS

1. Neurotoxin (paralytic , excitatory).
2. Myotoxins (systemic or local myolysis).
3. Cardiotoxins.
4. Haemostasis system toxins (bleeding , thrombosis).
5. Haemorrhagic toxins (vessel damage, fluid extravasations, blistering, echymosis, shock)
6. Nephrotoxin.
7. Necrotoxin.
8. Allergic toxin.

Local and systemic effects of envenomation

Local effects

Pain

Sweating

Erythema

Blistering

Necrosis

Swelling

Bleeding and bruising

Non specific systemic effects

Headache

Nausea , vomiting , diarrhoea

Abdominal pain, tachy or bradycardia, hypo or hypertension

Pulmonary edema, dizziness, shock, convulsions , cardiac arrest.

Specific systemic effects

Neurotoxic flaccid paralysis, excitatory neurotoxicity, rhabdomyolysis, coagulopathy, cardiotoxicity, acute kidney injury.

GENERAL APPROACH TO THE ENVENOMED PATIENT



First aid :

- ? Keep the patient still, immobilize the bitten limb with a splint, support vital system especially airway.
- ? Do Not use pressure bandage or tourniquet except for certain types (Myanmar, New Guinea, Australia). Tourniquets have the potential to cause catastrophic distal limb injuries in snake bite, when applied too narrowly, too tightly, or left on too long
- ? Avoid (cut & suction of the bitten site, suction device, local chemicals, snake stones,

GENERAL APPROACH TO THE ENVENOMED PATIENT



Transporting patients :

- ❓ When possible transport should be brought to the patient.
- ❓ Avoid any delay to medical facility.
- ❓ Support position of the patient during transport
- ❓ En route management of life-threatening conditions , as shock, or respiratory failure.

ADAM.



SNAKEBITE FIRST AID

Help yourself and others by following these first-aid steps immediately in a snakebite emergency and before reaching a health facility.

DO

DO stay calm and reassure the bitten person.



DO move slowly away from the snake.



DO rinse venom in the eyes with running water in case of spitting snakes.



DO leave the wound area (or bite mark) alone.



DO remove all tight items around the affected area.



DO lay the patient on his/her side and reduce movement of the affected area.



DO rush to the nearest health facility for medical treatment.



DON'T

DON'T panic. Instead, calmly follow these first aid steps.



DON'T attack or kill the snake. If you are close enough to hurt it, it is close enough to defend itself by biting you.



DON'T rub the eyes. Rubbing causes irritation to the eye and spreads the venom.



DON'T wash, cut or suck the wound. It can lead to infections and venom spreading.



DON'T tie the affected area to stop blood circulation. It can lead to infections that cause loss of limbs or even death.



DON'T lay the patient on his/her back. Lying on the back can block the airways.



DON'T use traditional methods or any unsafe treatments. For venomous snakebites, blackstones and herbs don't help and will delay medical treatment.



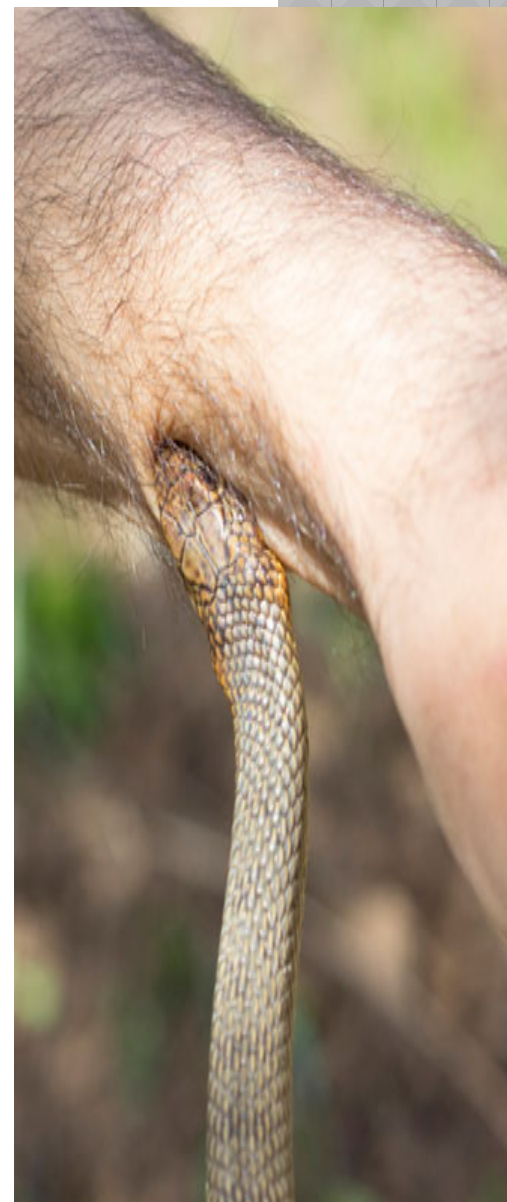
Snakebite EMERGENCY?

In case of a snakebite emergency, go to a health facility **IMMEDIATELY!**

8 SIMPLE STEPS TO AVOID SNAKEBITES

- 1 Wear closed shoes or boots and watch your step in the bush.
- 2 Use a torch outside at night.
- 3 Keep food, water and livestock outside your house. Store food and water in closed containers and livestock outside the house to prevent rats and mice entering the house.
- 4 Store firewood distant from your house. Store firewood distant from buildings to prevent snakes hiding and coming into the building.
- 5 Sleep off the ground and use mosquito nets to avoid snakebites during sleep.
- 6 Seal holes in your walls to prevent snakes entering the house.
- 7 Cut grass and clear the ground around your house. Clearing the area gets rid of hiding places for snakes.
- 8 Find out how to get to the nearest health facility with antivenom.

This poster is an initiative of Health Action International (HAI) and Global Snakebite Initiative (GSI).
HAI: www.haiweb.org/what-we-do/snakebite-envenoming
GSI: www.snakebiteinitiative.org
To order posters, email: haiglobal@haiweb.org or download from www.haiweb.org.



Highly Venomous



Thick tail, thin pincers. Intense pain at sting site. May be life threatening. Expect Class 2 and 3 Symptoms.

Mildly Venomous



Medium tail, medium pincers. Wasp sting-like symptoms causing localised pain. Expect Class 1 and 2 Symptoms.

Weakly Venomous



Thin tail, thick pincers. Pin prick-like symptoms with itching. Expect Class 1 Symptoms.

First Aid Techniques

Class 1 and 2 Scorpion Envenomation

- Calm the patient.
- Alert security or safety officer, arrange for transportation.
- Make the patient comfortable.
- Transport the patient to professional medical assistance.

Class 3 Scorpion Envenomation

- Calm the patient.
- Alert security or safety officer, arrange for transportation.
- Closely monitor vital signs.
- Transport the patient to professional medical assistance.
- Antivenom may be required.

HOSPITAL MANAGEMENT

? History :

? When was the patient exposed to venomous bite/sting?

? Was the organism causing it seen? Size? Color?

? What were the circumstance ? On land? Water?

? How many bites?

? What was first aid ?? When started? For how long?

? What are the symptoms?

? Past medical history? Any history of previous allergy or exposure to antivenom/venom??

EXAMINATION , INVESTIGATIONS & TREATMENT

- ❓ Local and systemic examination , for identifying any life threatening conditions.
- ❓ Treatment includes (anti-venom, supportive treatment)